



SERVICE MANUAL

MULTI CHANNEL AM/FM TRANSCEIVER

TCB-770

TTI Tech

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SECTION1 SPECIFICATIONS

GENERAL	Channel	40 (See the frequency band chart)
	Frequency Range	26.96 MHz ~ 27.99125 MHz
	Operating mode	F3E (FM), A3E (AM)
	Frequency Control	PLL Synthesizer
	Frequency Tolerance	0.002%
	Operating Temperature Range	-10 to + 55 °C
	Microphone	Plug-in Type
	Input Voltage	13.2V DC $\pm$ 15%
	Size	140 (W) x 186 (L) x 37 (H)
	Weight	990 g
	Antenna Connector	Antenna Connector
TRANSMITTER	Power Output	Duty cycle 10% 4 Watts @13.8V DC
	Modulation	AM:from 85% to 95% FM:1.8KHz $\pm$ 0.2KHz
	Frequency Response	300Hz to 3000Hz
	Output Impedance	50ohms, Unbalanced
	Harmonic Suppression	More than -36dBm
	Current Drain	AM Full Mod. 1.6A Max.
RECEIVER	Receiving System	Dual conversion superheterodyne
	IF Frequencies	Double Conversion 1st 10.695MHz/2nd 455KHz
	Sensitivity	0.7 μ V for 10dB(S+N)/N in AM Mode 0.5 μ V for 20dB SINAD in FM Mode
	Audio Output Power	2.0W @ 8 Ohm
	Audio Distortion	Less then 8% @ 1KHz
	Image Rejection	60 dB
	Adjacent Channel Rejection	60 dB
	Conducted Spurious	More than 60 dB
	Frequency Response	300 to 2500Hz
	Built-in Speaker	8 Ohms, round
	Squelch	Adjustable; Threshold less than 1 microvolt

* Specification is subject to change without prior notice.

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1. General

1-1. Receiver

- ① Display : 40 Channels and other functions indication
- ② Frequency Range : 26.965 to 27.405 [MHz]
- ③ Frequency Response : 300 to 2,500 [Hz]
- ④ Power Source : 13.2 [V] DC
- ⑤ Audio Output Load : 8 [OHM]
- ⑥ Audio Output : 4.0 [W] (or More)
- ⑦ Squelch : Adjustable from 0.3[uV] to 100 [uV]
- ⑧ Sensitivity :
 - FM : 20 dB [SINAD] under 1.0 [uV] RF Signal or less
 - AM : 10 dB [S/N] under 1.0 [uV] RF Signal or less
- ⑨ Intermediate Frequency :
 - 1st IF : 10.695 [MHz]
 - 2nd IF : 455 [KHz]

1-2. Transmitter

- ① Carrier Power(Conducted) : AM 1 [W] / FM 4 [W]
 - ② Current Drain (13.2 [V] Supply Voltage)
 - No Modulation : 1,000 [mA]
 - Max Modulation : 1,400 [mA]
 - ③ Modulation Capabilities
 - AM : ± 90 [%]
 - FM : ± 1.8 [KHz/Dev]
 - ④ Spurious Radiation : Less than -72 [dBc]
 - ⑤ Antenna Impedance : 50 [OHM]
 - ⑥ Frequency Tolerance : 0.002 [%]
-

2. Technical Description

2-1. General

Model CB-770 is an mobile type AM/FM radio transceiver for use of the Citizen Radio Service.

- **Front Panel Controls**

- ① LCD (Cannel and RX/TX Signal Level Indicator).
- ② Receiver Audio Control (With Power ON/OFF Control) Volume
- ③ Squelch Control Volume
- ④ RF Gain Control Volume
- ⑤ MIC GAIN Control Volume
- ⑥ CHANNEL UP/DOWN Rotary Switch
- ⑦ EMG(Emergency Channel Select) Key
- ⑧ AM/FM Select Key
- ⑨ Scan Key
- ⑩ Q.CHANNEL UP Key
- ⑪ Q.CHANNEL DOWN Key

- **Accessorily Connectors**

- ① Antenna Socket
- ② External Speaker Jack
- ③ External Signal Meter Jack
- ④ Microphone(Condenser Microphone)

2-2. Type of emission : AM(A3E) , FM(F3E)

2-3. Frequency Table

Channel	Frequency (MHz)	Channel	Frequency (MHz)
1	26.965	21	27.215
2	26.975	22	27.225
3	26.985	23	27.235
4	27.005	24	27.245
5	27.015	25	27.255
6	27.025	26	27.265
7	27.035	27	27.275
8	27.055	28	27.285
9	27.065	29	27.295
10	27.075	30	27.305
11	27.085	31	27.315
12	27.105	32	27.325
13	27.115	33	27.335
14	27.125	34	27.345
15	27.135	35	27.355
16	27.155	36	27.365
17	27.165	37	27.375
18	27.175	38	27.385
19	27.185	39	27.395
20	27.205	40	27.405

2-4. RF Power Output

- AM : 1.0 [W]
- FM : 4.0 [W]

2-5. DC Input Voltage and Current with 13.2V DC Input to Power AMP

- Transmitter Power Amp : Voltage 12.5 V
Current 727 mA
- Transmitter Driver Amp : Voltage 12.5 V
Current 120 mA

2-6. Receiver IF and Local Oscillator Frequencies

- First IF : 10.695 [MHz]
 - Second IF : 455 [KHz]
 - First Local Oscillator : 10.695 [MHz] Below Receiving Frequency
 - Second Local Oscillation : 10.240 [MHz]
-

3. Semiconductors and Function

3-1. Transistor

Ref No.	Description	Manufacturer	Function
Q101	KTC3880S	KEC	RX Amplifier
Q102	KTK211	KEC	1'st Mixer
Q103	KTK211	KEC	1'st Mixer
Q104	KRC101S	KEC	RX Band1 Switching
Q105	KRC101S	KEC	RX Band2 Switching
Q121	KRC110S	KEC	
Q122	KRC112S	KEC	RX RSSI Level Control
Q151	KTC3875S	KEC	RX FM AF Amplifier
Q152	KRC112S	KEC	FM AF Mute at AM Mode
Q153	KRC101S	KEC	FM AF Mute at AM Mode
Q154	KRC112S	KEC	RX AF Mute at TX Mode
Q161	KTC3880S	KEC	AM IF Amplifier
Q162	KTC3880S	KEC	AM IF Amplifier
Q163	KRA102S	KEC	RX AM B+
Q164	KTA1504S	KEC	Automatic Noise Level Control
Q181	KTC3875S	KEC	RX RF Mute at TX Mode
Q182	KTC3875S	KEC	AGC
Q183	KTC3875S	KEC	AGC
Q201	KRC101S	KEC	AF Mute
Q202	KRC111S	KEC	AF Path
Q233	KTA1504S	KEC	Automatic Level Control
Q234	KTC3875S	KEC	Automatic Level Control
Q301	KTC3880S	KEC	VCO
Q302	KTC3880S	KEC	RX/TX VCO Buffer
Q303	KTC3880S	KEC	VCO Buffer
Q305	KRC101S	KEC	TX VCO Control
Q311	KRC101S	KEC	VCO Band1 Switching
Q312	KRC101S	KEC	VCO Band2 Switching
Q313	KRC101S	KEC	VCO Band3 Switching
Q502	KRC111S	KEC	FM TX AF Mute at AM TX
Q701	KTC3880S	KEC	TX Doubler
Q702	KTC3880S	KEC	TX Pre-Amplifier
Q703	KTC1006	KEC	TX Driver Amplifier
Q704	2SC2078	SANYO	TX Power Amplifier
Q742	KTC3875S	KEC	TX RF Mute at RX Mode
Q801	KRC104S	KEC	CPU Reset
Q802	KRC101S	KEC	CPU RX/TX Control
Q803	KRA101S	KEC	MIC Lock Control
Q911	KTA1658	KEC	Main Regulator
Q912	KRA1504S	KEC	Main Regulator
Q913	KRC112S	KEC	Main Regulator Control
Q914	KRC102S	KEC	Power Low Control
Q921	KTC3875S	KEC	8.2V Regulator
Q923	KTC3875S	KEC	RX B+
Q924	KRA1504S	KEC	TX B+
Q931	KTC3875S	KEC	Main Regulator[5.6V]
Q941	KRC111S	KEC	Mute

Q951	KTA1241	KEC	Back Light LED Control
Q952	KTA1241	KEC	Back Light LED Control
Q953	KRC101S	KEC	Back Light LED Control
Q954	KRC101S	KEC	Back Light LED Control
Q961	KRC110S	KEC	TX AF Mute at RX Mode
Q962	KRC111S	KEC	MIC AF Path at AM Mode

3-2. IC

Ref No.	Description	Manufacturer	Function
IC101	TK144889V	TOKO	IF IC
IC102	KIA358F	KEC	AGC, Squelch Control
IC201	KIA7217AP	KEC	AF Power Amplifier
IC301	LC7152NM	SANYO	PLL
IC401	KIA4558F	KEC	FM TX AF
IC801	TMP87PP21DF	TOSHIBA	CPU
IC802	24WC02	ATMEL	EEPROM

3-3. Manufacture Information

- KEC ----- KEC Co., LTD.
- SANYO ----- SANYO Semiconductor Co., LTD.
- TOKO ----- TOKO, Inc.
- TOSHIBA ----- Toshiba Semiconductor. Co., LTD.
- ATMEL ----- ATMEL Co., LTD

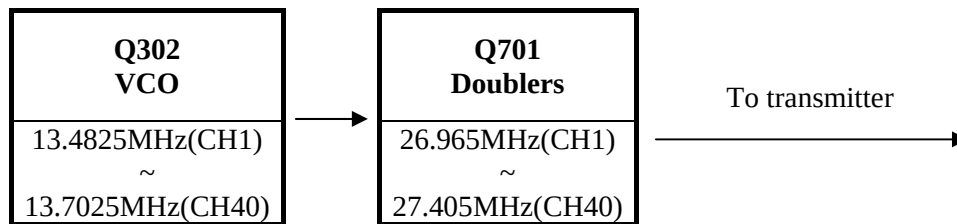
4. Description of Frequency Determining and Stabilizing Circuit

4-1. Introduction

The Frequencies for transmitter and receiver first local frequencies are all derived from a signal 4.5MHz crystal by means of a phase locked loop(PLL).

The first local oscillator frequencies are 16.270MHz(CH1) to 16.710MHz(CH40). The second local frequency is fixed at 10.240MHz to generate second IF 455KHz.

Transmit, the VCO of the PLL operates 13.4825MHz(CH1) to 13.7025MHz(40CH). The VCO frequency goes to the doublers circuit Q701, L701, L702 with doubles the frequency to generate 26.965MHz(CH1) to 27.405MHz(40CH).



The VCO operating frequency for the receiver is 16.27MHz(CH1) to 16.710MHz(CH40) as the first local oscillator, injected through the buffer amplifier Q302 into the first FET balanced mixer Q102, Q103.

4-2. Basic Synthesis Scheme

The crystal frequency(4.5MHz) is divided by 1800 times to make 2.5KHz which is fed to one side of the phase detector. The VCO output is divided by a programmable divider, and fed to other side of the phase detector Pin 9, 10 of IC301. Passing the phase detector output closes the feedback loop through an active low pass filter and using the output to control the VCO frequency through varicap diode D301. Under locked conditions, both of phase detector input signal must be identical at 2.5KHz.

The VCO frequency is then given by ;

$$\begin{array}{ll} \text{Receiver} & : F_{vco} / N = 0.005 \text{ [MHz]} \quad \text{OR} \quad F_{vco} = 0.005 \times N \text{ [MHz]} \\ \text{Transmitter} & : F_{vco} / N = 0.0025 \text{ [MHz]} \quad \text{OR} \quad F_{vco} = 0.0025 \times N \text{ [MHz]} \end{array}$$

Since “N” is an integer, the VCO frequency can be stepped up with in receiver mode 5KHz and transmitter mode 2.5KHz increments.

By suitable choice of “N” the desired output frequency can be obtained.

	Channel 1		Channel 40	
	N	Fvco	N	Fvco
Transmit	5393	13.4825	5481	13.7025
Receiver	3254	16.2700	3342	16.7100
	See the [Table 1]for other Channels			

The VCO frequency goes to the doublers circuit, with doubles the incoming signals.

	VCO Output Frequency	Doublers Output Frequency
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Transmit	CH 1 , 13.4825 MHz	26.965 MHz

Transmit	CH 40 , 13.7025 MHz	27.405 MHz

Since all frequencies are obtained from the crystal controlled PLL oscillator, all outputs are coherent with the crystal oscillator frequency and matching the sample percentage accuracy.

Note that the reference frequency of 5KHz(receiver) and 2.5KHz(transmitter) is obtained by dividing the 4.5MHz by 900 times and 1800 times

See Table 1 for transmitter/receiver mode VCO frequencies.

Program Data & Frequency[Table 1]

Channel	Frequency [MHz]	RX		TX	
		N	Fvco	N	Fvco
1	26.965	3254	16.27	5393	13.4825
2	26.975	3256	16.28	5395	13.4875
3	26.985	3258	16.29	5397	13.4925
4	27.005	3262	16.31	5401	13.5025
5	27.015	3264	16.32	5403	13.5075
6	27.025	3266	16.33	5405	13.5125
7	27.035	3268	16.34	5407	13.5175
8	27.055	3272	16.36	5411	13.5275
9	27.065	3274	16.37	5413	13.5325
10	27.075	3276	16.38	5415	13.5375
11	27.085	3278	16.39	5417	13.5425
12	27.105	3282	16.41	5421	13.5525
13	27.115	3284	16.42	5423	13.5575
14	27.125	3286	16.43	5425	13.5625
15	27.135	3288	16.44	5427	13.5675
16	27.155	3292	16.46	5431	13.5775
17	27.165	3294	16.47	5433	13.5825
18	27.175	3296	16.48	5435	13.5875
19	27.185	3298	16.49	5437	13.5925
20	27.205	3302	16.51	5441	13.6025
21	27.215	3304	16.52	5443	13.6075
22	27.225	3306	16.53	5445	13.6125
23	27.255	3312	16.56	5451	13.6275
24	27.235	3308	16.54	5447	13.6175
25	27.245	3310	16.55	5449	13.6225
26	27.265	3314	16.57	5453	13.6325
27	27.275	3316	16.58	5455	13.6375
28	27.285	3318	16.59	5457	13.6425
29	27.295	3320	16.60	5459	13.6475
30	27.305	3322	16.61	5461	13.6525
31	27.315	3324	16.62	5463	13.6575
32	27.325	3326	16.63	5465	13.6625
33	27.335	3328	16.64	5467	13.6675
34	27.345	3330	16.65	5469	13.6725
35	27.355	3332	16.66	5471	13.6775
36	27.365	3334	16.67	5473	13.6825
37	27.375	3336	16.68	5475	13.6875
38	27.385	3338	16.69	5477	13.6925
39	27.395	3340	16.70	5479	13.6975
40	27.405	3342	16.71	5481	13.7025

4-3. Descriptions of Each Block

4-3-1. Introduction

The synthesizer is implemented with the following components;

PLL IC (IC301)
X-TAL (X301)
VCO, VARICAP DIODE (D301)

IC1301 is CMOS LSI that includes most of PLL block.

The Q301, L301, C311, C312, C313, C314, variable capacitance diode D301 are clapping oscillator circuit to operate as a VCO of the IC301. Q305 is a switching transistor to connect or disconnect the tuning capacitor in the VCO oscillator tank circuit for transmitter or receiver.

Q302 works as a buffer amplifier for RX local frequencies (16MHz) and TX carrier generating frequencies(13MHz)

4-3-2. Reference Frequency

The crystal X301(4.5MHz) and other components at Pin 1 and 24 of IC301 can make a reference frequency oscillator with internal amplifier.

4-3-3. VCO

Q301 and surrounding parts are consisting a clapping oscillator works as a VCO of IC301. With appropriate control voltage on D301 the VCO can be oscillate over the required range of 13.4825MHz to 16.710MHz.

4-3-4. Programmable Divider and its Control

The programmable input for each channel are stetted by the PLL Clock(Pin 11), PLL Data(Pin 12), PLL Enable(Pin 13) of IC801. Each input signal to control the PLL IC is done with provide key input Pin 18, 19, 25, 26, 27, 28, 29, 32, 33, 34, 35, 37. For each key input, an internal code converts EEPROM appropriate control to the programmable divider for that channel.

Since the change transmit and receive, and additional bit is required at Pin 8 of IC801 to allow the ROM to recognize the status TX or RX.

During transmit the push to talk switch makes Pin 8 ground with BRT(Q802), PLL IC works under transmit status.

The programmable divider output fed to the phase detector for compare with the 2,5KHz reference frequency IC301. See table 1 for actual input and divide ratio on all channels.

4-3-5. Phase Detector and VCO Control

The phase detector is a digital phase comparator witch compares the phase of the reference signal with programmable divider output square waves and develops a series of pulses whose dc level depends on the phase error of each signal.

The phase detector pulse output is fed to an active low pass filter and RC low pass filter output signal of IC301 is filtered and fed to varicap D301 control the VCO frequency.

4-3-6. Transmitter / Receiver Buffer AMP

Output signal of Q301 is fed into the buffer amplifier Q302, L302 to generate TX carrier frequency and 1'st local frequencies.

4-3-7. Transmitter Doublers

The output signal of Q302, L302 goes to an amplifier with tuning circuit Q701, L701, L702 which doubles incoming 13MHz signals.

4-3-8. Switching of Turning Capacitor in VCO

The VCO circuit must tune with a wide range of frequencies 13.4825MHz ~ 13.7025MHz for transmitter and 16.27MHz ~ 16.71MHz for receiver.

To comply above range of VCO, the tuning capacitance should switch for transmission. The tuning circuit consists with L301, C311, C312, C313, and C314. When the VCO is working as a receiver Q305 becomes turn OFF. So, L301 and C311, D301 makes turning function.

When transmitting, Q305 becomes ON. So, L301 and parallel capacitance of C316 and C311 make turning function.

4-3-9. Receiver Local Oscillator Outputs

- FIRST MIXER :

The secondary output signal of L107 is injected to the sources of 1st mixer Q102, Q103 in the 1st IF mixer section.

- SECOND MIXER :

The output of 10.24MHz oscillator circuit with X101 is injected into the IF IC(IC101) internally. Incoming IF signal and 10.24MHz signal are mixed inside the IF IC to extract 2nd IF signal 455KHz.

FM audio Signals are recovered with the way of quadrature detector.

AM signals are recovered with envelope detector.

4-4. Frequency Stability

Let : F_o = Crystal oscillator frequency

F_r = Phase detector reference frequency

F_{vco} = VCO frequency

F_t = Transmit frequency

Then : Receiver : $F_r = F_o/900$

Transmitter : $F_r = F_o/1800$

And under locked conditions : $F_r = F_{vco} / N$

Where, "N" is the programmable divider divide ratio.

Then : $F_{vco} = N \times F_r$

From which it can be seen, the percentage error in F_t is the same as the percentage error in F_o . The stability of the crystal oscillator is determined primarily by the crystal itself and having lesser deviation by the active and passive components of the oscillator. The choice of crystal and components is such that the required frequency stability is maintained over the required voltage and temperature range.

4-5. Description of other Circuits

4-5-1. Transmitter

A. RF Amplification

RF carrier frequencies are obtained at the output of doublers amplifier Q701 and turning IFT coil L701, L702. The input of VCO frequencies 13MHz is selected at the doublers output, which creates the necessary 27MHz. The output of doublers amplifier Q701 is fed through turning IFT coil L701, L702 to the base of pre driver amplifier Q702. It's output; 27MHz is coupled to RF driver amplifier Q703, a low-level class C power amplifier. Driver Q703 supplies the necessary power gain to operate RF final Q704 at the maximum efficient. The output of Q704 is supplied to the antenna through L-C turning circuit.

B. Circuit for Suppression of Spurious Radiation

The turning circuit between the output of final amplifier Q704 and antenna, 4-stage " π " network C722, C723, C724, L711, C725, C726, L712, C729, C731, L732, C732, C733, C734, L714, C735 serves as a spurious radiation suppressor. This network also servers to match the impedance between TX power amplifier Q704 and the antenna.

C. Circuits for Limiting Power

After finished all alignment, the constant voltage supply circuit limits the available power 4W or slightly less. RV911 and corresponding three-transistor control supply voltage of RF power amplifier. When power low switch function Q914 changed the supply voltage. Tune all the trimming parts for maximum indication of RF power meter and adjust RV571 to make 4W indication of RF power meter. After finishing the above adjustments check the RF power meter reading is changed 1W under "LOW" state.

The turning is adjustment so that the actual power is from 3.8W to 4.0W. There are no other additional controls for adjusting the TX output power.

D. Modulation Control

Modulation of the RF is a process that begins with the audio picked up by the microphone.

<FM>

The microphone input is fed to mic audio amplifier IC401 that drives modulation vricap diode D302 in the VCO circuit RV401 limits the incoming modulation audio levels to inhibit over modulation. While reading the modulation factor on the modulation analyzing equipment, adjustment RV401 shall not exceed 2.0KHz deviation.

<AM>

Modulation signals are filtered with RC network and goes to the audio power amplifier IC201 in to make normal signal level to achieve wanted modulation. IC201 drives T201, which is a combination AF output/modulation transformer. During transmit, one of the secondary windings of T201 is tied between the 13.2V DC supply and the collectors of RF driver Q703 and RF power amplifier Q704. As the audio passes through T201 it causes the collector supplies of Q703 and Q704 to vary with audio, producing an AM signal at carrier frequency.

To avoid over modulation of the carrier, a protection of the modulating signal is fed back from the T201 through the AMC circuit, Q233 and Q234, to control the gain of IC201 and sets the maximum level of modulation. Form the center tab of RV231 and Q233 the feedback signal is rectified by diode D233, filtered by C233, and supplied the Q234. The collector of Q234 is tied directly TX audio input to control gain. That is, when the audio output is higher that the preset level of RV231, the information is reflected through Q233 and Q234, reducing the gain of IC201

4-5-2. Receiver

Overload protection is provided to receiver's semiconductors by diode D101. These diodes have a little effect on the incoming signal from another CB station; they protect the receiver from stray transmitter energy.

In the receiver mode of operation, Q923 transistor is turned on. Also bias voltage is applied to Q101, Q102, Q103, Q106.

Q101 is a 27MHz RF input amplifier, and any excessive input signal is limited by diode D101.

CB receiver is dual conversion super-heterodyne type with the first IF 10.695MHz and the second IF 455KHz.

Receiver is separated to blocks, 1'st IF section and 2'nd IF section. The PLL synthesizer supplies 1'st local frequency 16.270MHz ~ 16.710MHz. The amplified 27MHz is mixed. With the provide 1'st local frequencies Q102, Q103 mixes the incoming RF signal to generate 1'st IF signal. The resulting first IF is 10.695MHz. Q102 and Q103 is the first converter, and 10.695MHz is sharply filtered by L108 and crystal filter CF101. The first IF is again mixed with a second local oscillator of 10.24MHz.

With the 10.240MHz signal, IC101 FM IF IC converts the incoming signals to generate 2'nd IF signal and recovered audio signals. 2'nd IF is filtered by a razor sharp ceramic filter CF101 coupled. The 455KHz signal from the 2'nd IF filter was amplified and limits internally. After amplification the signals fed to the quadrature detector loop L109. Then could see the recovered signals Pin 11 of IC101.

With the amplitude of recovered signals, Q151 serves as an amplifier.

AM signals from the Pin 6 of IC101 were tapped with C161 and amplified two-stage amplifier Q161, Q162. Q161 is a first 455KHz amplifier, and the Q161 being the last amplifier. D106 is a detector diode which produce audio signal as well as negative DC voltage for AGC action. The negative voltage also provides forward biasing to the emitter of ANL clipping transistor Q164. The biasing voltage has a time constant determined R174 and C176. Therefore any sharp negative going pulse from D161 will back bias Q164 and clipped. The way to recover the AM information audio is envelope detector.

5. Test Equipment Setup and Alignment Instructions

5-1 General Section

5-1-1 Test Equipment Required

- DC power supply(13.2V/3A)
- DC Voltmeter or Oscilloscope
- RF attenuator (30dB)

5-1-2 Alignment Procedure

Step	Setting	Connection	Adjuster	Adjust for
1	RX VCO voltage Adj. Channel & Frequency : 1CH, 26.956 MHz Mic : Receive Volume : Optional Squelch : Optional RF Gain : Optional Mic Gain : Optional	DC voltmeter to VCO Test point (Figure 1)	L301	1.5~2.0 V DC
2	TX VCO voltage Adj. Channel & Frequency : 1CH, 26.956 MHz Mic : Transmit(No Mod) Volume : Optional Squelch : Optional RF Gain : Optional Mic Gain : Optional	DC voltmeter to VCO Test point (Figure 1)	L301	1.5~2.0 V DC

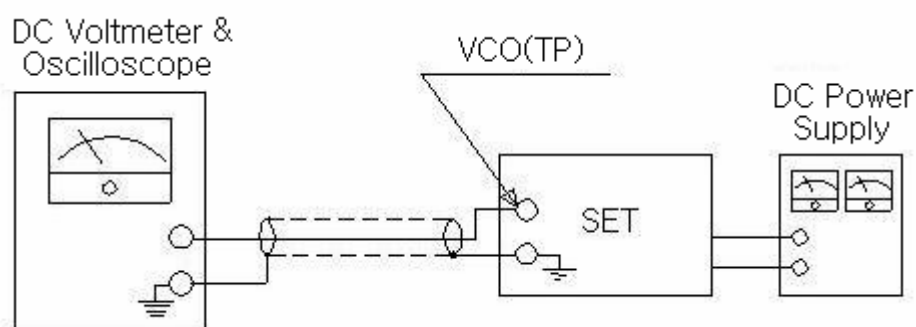


Figure 1

5-2 Transmitter Section

5-2-1 Test Equipment Required

- RF Power Meter
- 50 ohms load (non-inductive)
- RF attenuator (30dB)
- Oscilloscope
- Audio generator
- DC power supply(13.2V/3A)
- Spectrum analyzer
- Frequency counter
- Coupler
- Modulation meter

5-2-2 Alignment Procedure

Step	Setting	Connection	Adjuster	Adjust for
1	RF Power Adj. Channel : 19CH Function : AM or FM Mode CB/ANL Off/DX MIC : Transmit Volume : Optional Squelch : Optional RF Gain : Optional MIC Gain : Minimum	Connect dummy load and RF Power Meter to the EXT-ANT connector on the set(Figure 3)	L302 L701 L702 L703	Maximum indication on the Power Meter(4.0W). If indication is not in 4W range, ACP Adjustment(RV911)
2	Frequency Adj. Channel & Frequency : 19CH, 26.956 MHz Function : AM or FM Mode CB/ANL Off/DX Mic : Transmit(No Mod) Volume : Optional Squelch : Optional RF Gain : Optional MIC Gain : Optional	Connect dummy load and Frequency Counter through Coupler to RF Power Meter. Connect RF Power Meter to EXT-ANT connector on the set(Figure 2).	CT302	Be sure that the indication of the transmitter frequency is 27.185MHz $\pm$ 300 Hz on the Frequency Counter
3	AM Modulation Adj. Channel & Frequency : 19CH, 27.185MHz MIC: Transmit Function : AM Mode CB/ANL Off/DX Volume : Optional Squelch : Optional RF Gain : Optional MIC Gain : Max	Connection the audio generator (set to 1 KHz)to the microphone. Connect the modulation meter through the RF attenuator to the ANT Connector. Adjust the audio signal level to obtain by 50% modulation. When you increase the audio signal by 20 dB, the modulation should not exceed 95% Modulation(Figure 3).	RV231	85% ~ 95%

4	FM Deviation Adj. Channel & Frequency : 19CH, 27.185MHz MIC : Transmit Function : FM Mode CB/ANL Off/DX Volume : Optional Squelch : Optional RF Gain : Optional MIC Gain : Max	Connection the audio generator (set to 1 KHz)to the microphone. Connect the modulation meter through the RF attenuator to the ANT Connector. Adjust the audio signal level to obtain by 1KHz deviation. When you increase the audio signal by 20 dB, the deviation should not exceed 2.0KHz deviation(Figure 3).	RV401	1.8KHz ~ 2.0KHz
5	TX Signal Meter Adj. Channel & Frequency : 19CH, 27.185MHz MIC : Transmit Function : FM Mode CB/ANL Off/DX Volume : Optional Squelch : Optional RF Gain : Optional MIC Gain : Minimum	Connected through Dummy load to RF Power Meter. Connect RF Power to EXT-ANT connector on the set(Figure 2)	RV571	Adjust for 4 watts RF output power on the signal meter.

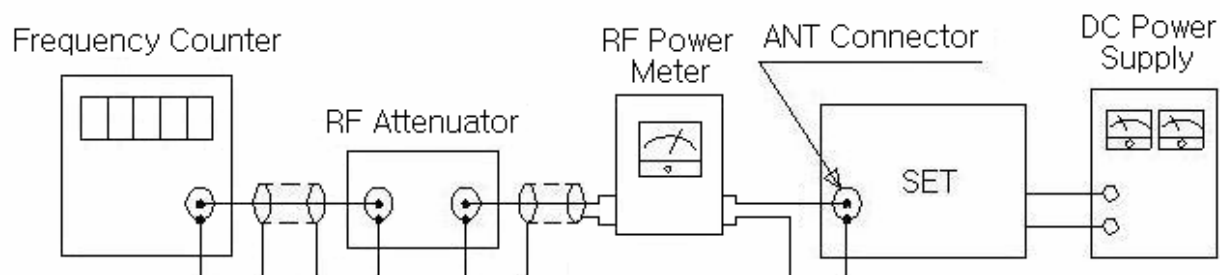


Figure 2

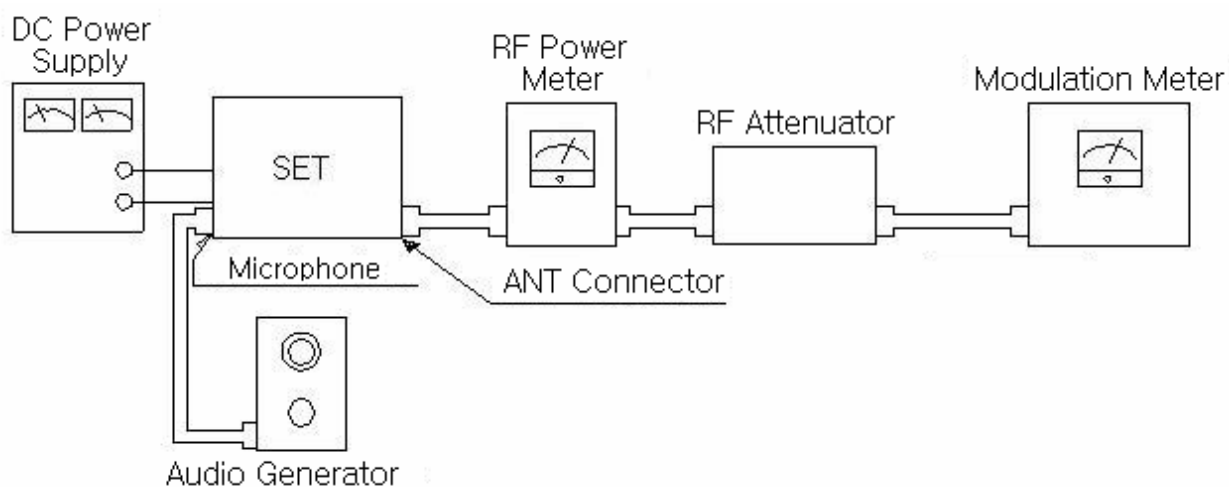


Figure 3

5-3 Receiver Section

5-3-1 Test Equipment Required

- Standard Signal Generator(SSG)
- DC power supply(13.8V/3A)
- AC Level Meter
- Distortion Meter
- Oscilloscope
- SINAD Meter
- 8 ohm Dummy Load

5-3-2 Alignment Procedure

Step	Setting	Connection	Adjuster	Adjust for
1	AM Sensitivity Adj. Channel & Frequency : 19CH, 27.185MHz MIC : Receive Function : AM Mode CB/ANL Off/DX SSG : 27.185MHz, 1KHz, -67dBm, 30% Mod. Volume : Optional Squelch : Minimum RF Gain : Maximum	Connect RF Signal Generator to EXT-ANT Connector. Connect AC Level Meter and Distortion Meter and SINAD Meter across EXT SPK jack with 8 ohm Dummy Load(Figure 4)	L101 L103 L104 L105 L107 L108 L112	Audio Maximum, Distortion Minimum. Reduce output from SSG until the audio output becomes about 10dB S/N
2	FM Sensitivity Adj. Channel & Frequency : 19CH, 27.185MHz MIC : Receive Function : FM Mode CB/ANL Off/DX SSG : 27.185MHz, 1KHz, -67dBm, 1.5KHz Dev. Volume : Optional Squelch : Minimum RF Gain : Maximum	Connect RF Signal Generator to EXT-ANT Connector. Connect AC Level Meter and Distortion Meter and SINAD Meter across EXT SPK jack with 8 ohm Dummy Load(Figure 4)	L109	Audio Maximum, Distortion Minimum. Reduce output from SSG until the audio output becomes about 20dB SINAD
3	Squelch Adj. Channel & Frequency : 19CH, 27.185MHz MIC : Receive Function : FM Mode CB/ANL Off/DX SSG : 27.185MHz, 1KHz, -67dBm, 1.5KHz Dev. Volume : Optional Squelch : Maximum RF Gain : Maximum	Connect RF Signal Generator to EXT-ANT Connector. Connect AC Level Meter and Distortion Meter and SINAD Meter across EXT SPK jack with 8 ohm Dummy Load(Figure 5)	RV131	Adjust until the audio output appears.

4	RX Signal Meter Adj. Channel & Frequency : 19CH, 27.185MHz MIC : Receive Function : FM Mode CB/ANL Off/DX SSG : 27.185MHz, 1KHz, -62dBm, 1.5KHz Dev. Volume : Optional Squelch : Maximum RF Gain : Maximum	Connect RF Signal Generator to EXT-ANT Connector. Connect AC Level Meter and Distortion Meter and SINAD Meter across EXT SPK jack with 8 ohm Dummy Load(Figure 4)	RV572	Adjust for S-9 on the signal meter.
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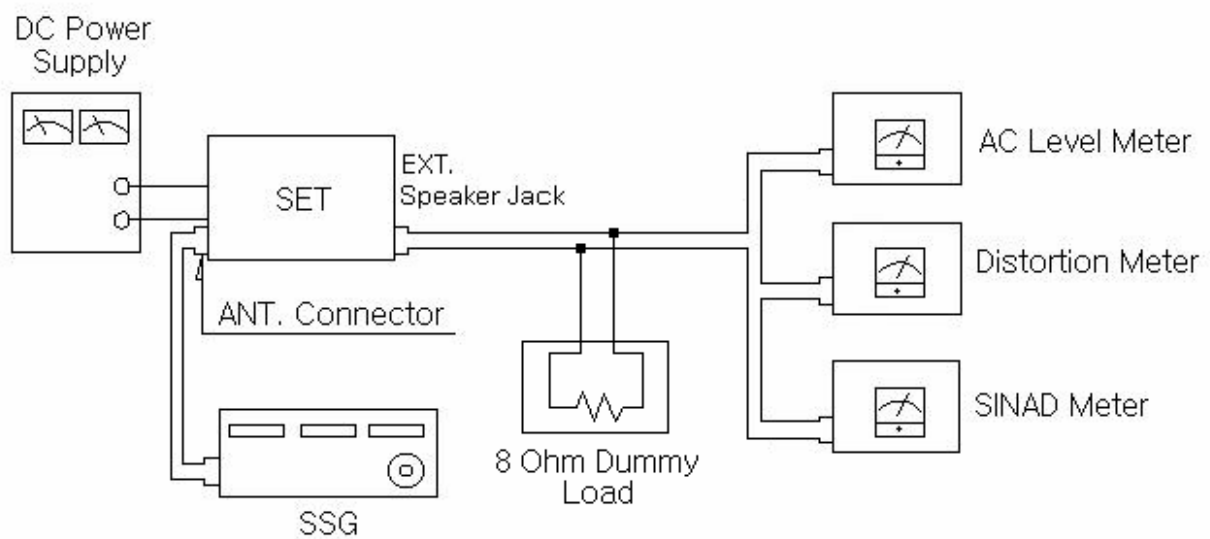


Figure 4

SECTION3 PART LIST

ITEM	NAME & DESCRIPTION	VENDOE CODE	MAKER	VENDOR	ASS'Y	Q'ty	Location NO	REMARKS
* ELECT Parts								
	COVER ASS'Y							
1	SPEAKER	66mm	66AR000-8P-4K11	ASAHI	ASAHI	MAN	1	SPK201
2	WIRE HARNESS	PH-2PIN TO OPEN	UL1007 #26 L=200mm	FINE	FINE	MAN	1	J204
	ESCUTCHEON ASS'Y							
3	PCB VOLUME	1LAYER		CAPITAL		MAN	1	
4	VR	B10K/A10K	SJ093D 15H 10F B10K+SA10K	SEJIN	SEJIN	MAN	1	VR101
5	VR	B50K/B50K	SJ093D 15H 10F B50K/B50K	SEJIN	SEJIN	MAN	1	VR102
6	FILM RESISTOR	10K	1/8W 5% ST			MAN	1	R400
7	SW ROTARY		SJ-PS2101 20SK	SEJIN	SEJIN	MAN	1	SW811
8	COIL AXIAL	6.8UH	LAL03TB6R8K			MAN	4	L501, L502, L504, L506
9	WAFER	16PIN	PH-16A 180C	FINE	FINE	MAN	1	J101
10	WIRE HARNESS	PH-16PIN*2	UL1007 #26 L=50MM	FINE	FINE	MAN	1	J101 --- J102
11	WIRE HARNESS	PH-6PIN TO OPEN	UL1007 #26 L=70MM	FINE	FINE	MAN	1	J502
12	CONNECTOR SOCKET		N-16-6(RPCB)-M	SAMWOO		MAN	1	J501
13	HEAT SINK TUBE		diameter 4 mm x 15mm			MAN	6	ECO050321-01
	LCD PCB ASS'Y							
14	PCB LED	1 LAYER		CAPITAL		MAN	1	
15	PCB DISPLAY	FR-4 / R-1		CAPITAL		MAN	1	ECO050809-01
16	CHIP RESISTOR	1K	1/16W 5% T 1608	PHILIPS		SMD	1	R810
17	CHIP RESISTOR	10K	1/16W 5% T 1608	PHILIPS		SMD	5	R805, R806, R807, R811, R814
18	CHIP RESISTOR	100K	1/16W 5% T 1608	PHILIPS		SMD	1	R804
19	CHIP RESISTOR	1M	1/16W 5% T 1608	PHILIPS		SMD	1	R803
20	CHIP RESISTOR	2.2K	1/16W 5% T 1608	PHILIPS		SMD	5	R832, R833, R834, R835, R836
21	CHIP RESISTOR	220K	1/16W 5% T 1608	PHILIPS		SMD	1	R808
22	CHIP RESISTOR	47K	1/16W 5% T 1608	PHILIPS		SMD	11	R812, R816, R831, R851, R852, R853, R854, R857, R859, R862, R864
23	CHIP CERAMIC	0.01UF	1608	PHILIPS		SMD	2	C806, C807
24	CHIP CERAMIC	1UF	1608	PHILIPS		SMD	1	C805
25	CHIP CERAMIC	68PF	1608	PHILIPS		SMD	2	C801, C802
26	CHIP TANTAL	10uF/16V	A TYPE			SMD	1	C811
27	CHIP TRANSISTOR	KRA101S	SOT-23	KEC		SMD	1	Q803
28	CHIP TRANSISTOR	KRC101S	SOT-23	KEC		SMD	1	Q802
29	CHIP TRANSISTOR	KRC104S	SOT-23	KEC		SMD	1	Q801
30	LED BACK LIGHT	CHIP LED	IWS-164-BXWF	ITSWELL		SMD	8	D953, D954, D955, D957, D958, D959, D965, D966
31	IC EEPROM	AT24C02 (SOP-8)		ATMEL		SMD	1	IC802
32	IC CPU	TMP87PP21DF		TOSHIBA		SMD	1	IC801
33	LCD DISPLAY	SDM8A2299A	FOR TCB-770	SANTECH		MAN	1	LCD801
34	SW TACT	SKTS-1102A		SEKWANG	SEKWANG	MAN	5	SW801, SW802, SW806, SW807, SW808
35	RESONATOR	4.0MHz	ZTA4.00MG	D.TRON	D.TRON	MAN	1	X801

ITEM	NAME & DESCRIPTION	VENDOE CODE		MAKER	VENDOR	ASS'Y	Q'ty	Location NO	REMARKS
	MAIN BODY ASS'Y								
36	CONNECTOR	CH-239(A)-M		SAMWOO		MAN	1	J701	ECO050511-02
37	TRANSISTOR	KTA1658		KEC		MAN	1	Q911	
38	TRANSISTOR	2SC2078		SANYO	YANG WON	MAN	1	Q704	
39	IC	KIA7217AP		KEC		MAN	1	IC201	
40	POWER CODE		2PIN PLUG ASS'Y	FINE	FINE	MAN	1	J900	
	MAIN PCB AUTO ASS'Y								
41	PCB MAIN	FR-4 / R-1		CAPITAL		SMD	1		ECO050809-01
42	CHIP RESISTOR	0	1/16W 5% T 1608	PHILIPS		SMD	3	R108, R137, R201	ECO050412-01
43	CHIP RESISTOR	1	1/16W 5% T 1608	PHILIPS		SMD	1	R706	
44	CHIP RESISTOR	10	1/16W 5% T 1608	PHILIPS		SMD	2	R133, R301	
45	CHIP RESISTOR	100	1/16W 5% T 1608	PHILIPS		SMD	6	R104, R109, R134, R168, R314, R401	
46	CHIP RESISTOR	1K	1/16W 5% T 1608	PHILIPS		SMD	19	R126, R164, R182, R183, R187, R200, R222, R234, R238, R308, R312, R410, R501, R571, R572, R955, R956, R957, R958	ECO050412-01
47	CHIP RESISTOR	10K	1/16W 5% T 1608	PHILIPS		SMD	16	R123, R157, R161, R185, R315, R321, R322, R323, R324, R412, R413, R414, R703, R821, R823, R914	ECO050418-03 ECO050715-02
48	CHIP RESISTOR	100K	1/16W 5% T 1608	PHILIPS		SMD	9	R140, R142, R159, R186, R188, R202, R233, R309, R311	
49	CHIP RESISTOR	2M	1/16W 5% T 1608	PHILIPS		SMD	1	R405	ECO050422-02 ECO050715-02
50	CHIP RESISTOR	1.5K	1/16W 5% T 1608	PHILIPS		SMD	2	R742, R924	
51	CHIP RESISTOR	15K	1/16W 5% T 1608	PHILIPS		SMD	3	R155, R166, R235	
52	CHIP RESISTOR	150K	1/16W 5% T 1608	PHILIPS		SMD	1	R701	
53	CHIP RESISTOR	2.2	1/16W 5% T 1608	PHILIPS		SMD	1	R711	
53	CHIP RESISTOR	22	1/16W 5% T 1608	PHILIPS		SMD	1	R204	ECO050422-02
54	CHIP RESISTOR	220	1/16W 5% T 1608	PHILIPS		SMD	2	R169, R702	
55	CHIP RESISTOR	2.2K	1/16W 5% T 1608	PHILIPS		SMD	12	R131, R203, R237, R307, R402, R817, R822, R912, R922, R923, R961, R01	ECO050412-01
56	CHIP RESISTOR	22K	1/16W 5% T 1608	PHILIPS		SMD	6	R132, R162, R175, R302, R304, R305	
57	CHIP RESISTOR	220K	1/16W 5% T 1608	PHILIPS		SMD	2	R171, R177	
58	CHIP RESISTOR	2.2M	1/16W 5% T 1608			SMD	1	R153	
59	CHIP RESISTOR	2.7K	1/16W 5% T 1608	PHILIPS		SMD	1	R167	
60	CHIP RESISTOR	27K	1/16W 5% T 1608	PHILIPS		SMD	2	R102, R403	
61	CHIP RESISTOR	33K	1/16W 5% T 1608	PHILIPS		SMD	3	R172, R404, R407	
62	CHIP RESISTOR	3.3M	1/16W 5% T 1608	PHILIPS		SMD	1	R135	
63	CHIP RESISTOR	390	1/16W 5% T 1608	PHILIPS		SMD	1	R156	
64	CHIP RESISTOR	39K	1/16W 5% T 1608	PHILIPS		SMD	2	R163, R306	

ITEM	NAME & DESCRIPTION	VENDOE CODE		MAKER	VENDOR	ASS'Y	Q'ty	Location NO	REMARKS
65	CHIP RESISTOR	47	1/16W 5% T 1608	PHILIPS		SMD	3	R184, R705, R913	
66	CHIP RESISTOR	470	1/16W 5% T 1608	PHILIPS		SMD	7	R101, R105, R106, R165, R181, R236, R931	
67	CHIP RESISTOR	4.7K	1/16W 5% T 1608	PHILIPS		SMD	6	R138, R223, R406, R704, R915, R916, R962	ECO050412-01 ECO050418-03
68	CHIP RESISTOR	47K	1/16W 5% T 1608	PHILIPS		SMD	9	R122, R139, R174, R176, R300, R303, R409, R411, R415	ECO050715-02 ECO050812-02
69	CHIP RESISTOR	470K	1/16W 5% T 1608	PHILIPS		SMD	2	R154, R573	
70	CHIP RESISTOR	5.6K	1/16W 5% T 1608	PHILIPS		SMD	1	R408	
71	CHIP RESISTOR	56K	1/16W 5% T 1608	PHILIPS		SMD	2	R103, R313	
72	CHIP RESISTOR	6.8K	1/16W 5% T 1608	PHILIPS		SMD	1	R152	
73	CHIP RESISTOR	68K	1/16W 5% T 1608	PHILIPS		SMD	1	R189	
74	CHIP RESISTOR	82K	1/16W 5% T 1608	PHILIPS		SMD	1	R173	
75	THERMISTOR	100 OHM	NCP18XF101J03RB	MURATA		SMD	1	TH131	ECO050525-02
76	THERMISTOR	10K OHM	NCP18XH103J03RB	MURATA		SMD	1	TH101	
77	CHIP CERAMIC	0.001UF	1608	PHILIPS/SAMSUNG		SMD	4	C181, C230, C239, C404	
78	CHIP CERAMIC	0.01UF	1608	PHILIPS/SAMSUNG		SMD	38	C111, C114, C115, C116, C117, C118, C153, C154, C155, C156, C157, C161, C162, C183, C184, C201, C231, C234, C235, C323, C412, C415, C501, C502, C503, C504, C506, C703, C704, C708, C713, C901, C902, C903, C911, C922, C924, C932	
79	CHIP CERAMIC	0.1UF	1608	PHILIPS/SAMSUNG		SMD	23	C122, C123, C132, C134, C135, C137, C138, C168, C169, C180, C186, C205, C236, C308, C322, C325, C330, C401, C402, C405, C413, C571, C573	ECO050715-02
80	CHIP CERAMIC	1UF	1608	PHILIPS/SAMSUNG		SMD	2	C159, C961	
81	CHIP CERAMIC	0.022UF	1608	PHILIPS/SAMSUNG		SMD	5	C166, C203, C214, C238, C403	
82	CHIP CERAMIC	0.047UF	1608	PHILIPS/SAMSUNG		SMD	8	C152, C163, C164, C165, C212, C310, C721, C914	
83	CHIP CERAMIC	0.068UF	1608	PHILIPS/SAMSUNG		SMD	1	C410	
84	CHIP CERAMIC	10PF	1608	PHILIPS/SAMSUNG		SMD	3	C96, C113, C324	
85	CHIP CERAMIC	100PF	1608	PHILIPS/SAMSUNG		SMD	12	C101, C109, C303, C304, C305, C313, C314, C711, C712, C731, 733, C736	ECO050812-02
86	CHIP CERAMIC	150PF	1608	PHILIPS/SAMSUNG		SMD	4	C102, C104, C106, C311	
87	CHIP CERAMIC	16PF	1608	PHILIPS/SAMSUNG		SMD	1	C316	
88	CHIP CERAMIC	18PF	1608	PHILIPS/SAMSUNG		SMD	4	C98, C99, C112, C119	
89	CHIP CERAMIC	180PF	1608	PHILIPS/SAMSUNG		SMD	2	C701, C734	
90	CHIP CERAMIC	2PF	1608	PHILIPS/SAMSUNG		SMD	1	C730	
91	CHIP CERAMIC	22PF	1608	PHILIPS/SAMSUNG		SMD	2	C312, C727	
92	CHIP CERAMIC	220PF	1608	PHILIPS/SAMSUNG		SMD	6	C204, C211, C329, C717, C719, C724	
93	CHIP CERAMIC	3PF	1608	PHILIPS/SAMSUNG		SMD	1	C319	
94	CHIP CERAMIC	30PF	1608	PHILIPS/SAMSUNG		SMD	1	C315	
95	CHIP CERAMIC	33PF	1608	PHILIPS/SAMSUNG		SMD	3	C108, C302, C318	

ITEM	NAME & DESCRIPTION	VENDOE CODE		MAKER	VENDOR	ASS'Y	Q'ty	Location NO	REMARKS
96	CHIP CERAMIC	330PF	1608	PHILIPS/SAMSUNG		SMD	6	C714, C715, C723, C725, C726, C729	
97	CHIP CERAMIC	39PF	1608	PHILIPS/SAMSUNG		SMD	1	C705	
98	CHIP CERAMIC	47PF	1608	PHILIPS/SAMSUNG		SMD	7	C103, C105, C107, C182, C328, C720, C722	
99	CHIP CERAMIC	470PF	1608	PHILIPS/SAMSUNG		SMD	1	C718	
100	CHIP CERAMIC	56PF	1608	PHILIPS/SAMSUNG		SMD	1	C130	
101	CHIP CERAMIC	6PF	1608	PHILIPS/SAMSUNG		SMD	2	C317, C321	
102	CHIP CERAMIC	68PF	1608	PHILIPS/SAMSUNG		SMD	3	C97, C131, C732	
103	CHIP CERAMIC	680PF	1608	PHILIPS/SAMSUNG		SMD	1	C414	
104	CHIP CERAMIC	82PF	1608	PHILIPS/SAMSUNG		SMD	2	C406, C707	
105	CHIP CERAMIC	8PF	1608	PHILIPS/SAMSUNG		SMD	1	C326	
106	CHIP CERAMIC	9PF	1608	PHILIPS/SAMSUNG		SMD	1	C110	
107	CHIP TANTAL	2.2uF/16V	A TYPE			SMD	1	C301	ECO050418-03
108	CHIP TRANSISTOR	KTK211	SOT-23	KEC		SMD	2	Q102, Q103	
109	CHIP TRANSISTOR	KRA102S	SOT-23	KEC		SMD	1	Q163	
110	CHIP TRANSISTOR	KRC101S	SOT-23	KEC		SMD	10	Q104, Q105, Q153, Q222, Q305, Q311, Q312, Q313, Q953, Q954	ECO050412-01
111	CHIP TRANSISTOR	KRC102S	SOT-23	KEC		SMD	1	Q914	
112	CHIP TRANSISTOR	KRC110S	SOT-23	KEC		SMD	3	Q121, Q960, Q961	ECO050715-02
113	CHIP TRANSISTOR	KRC111S	SOT-23	KEC		SMD	3	Q202, Q502, Q962	ECO050412-01
114	CHIP TRANSISTOR	KRC112S	SOT-23	KEC		SMD	4	Q122, Q152, Q154, Q913	
115	CHIP TRANSISTOR	KTA1504S	SOT-23	KEC		SMD	4	Q164, Q233, Q912, Q924	
116	CHIP TRANSISTOR	KTC3875S	SOT-23	KEC		SMD	9	Q151, Q181, Q182, Q183, Q234, Q742, Q921, Q923, Q931	
117	CHIP TRANSISTOR	KTC3880S	SOT-23	KEC		SMD	8	Q101, Q161, Q162, Q301, Q302, Q303, Q701, Q702	
118	DIODE ZENER	Z02W5.1V	SOT-23	KEC		SMD	1	DZ181	
119	DIODE ZENER	Z02W5.6V	SOT-23	KEC		SMD	1	DZ931	
120	DIODE ZENER	Z02W8.2V	SOT-23	KEC		SMD	1	DZ921	
121	DIODE SI CHIP	KDS181	SOT-23	KEC		SMD	1	D401	
122	DIODE SI CHIP	KDS184	SOT-23	KEC		SMD	13	D102, D103, D105, D171, D180, D181, D233, D571, D572, D741, D742, D921, D961	ECO050412-01
123	DIODE SI CHIP	KDS226	SOT-23	KEC		SMD	1	D101	
124	DIODE VARICAP	KDV251S	SOT-23	KEC		SMD	2	D301, D302	
125	IC PLL	LC7152NM		SANYO	YANG WON	SMD	1	IC301	
126	IC OP AMP	KIA358F		KEC		SMD	1	IC102	
127	IC OP AMP	KIA4558F		KEC		SMD	1	IC401	
128	IC FM IF	TK14489V		TOKO		SMD	1	IC101	
	MAIN PCB MANUAL ASS'Y								
129	FILM RESISTOR	2.2	1/2W 5% ST MINI			MAN	1	R712	
130	FILM RESISTOR	10	1/2W 5% ST MINI			MAN	1	R921	

ITEM	NAME & DESCRIPTION	VENDOE CODE		MAKER	VENDOR	ASS'Y	Q'ty	Location NO	REMARKS
131	FILM RESISTOR	470	1/2W 5% ST MINI			MAN	2	R953, R954	ECO050418-03
132	FILM RESISTOR	4.7K	1/2W 5% ST MINI			MAN	1	R713	
133	FILM RESISTOR	1K	1W 5% ST MINI			MAN	2	R951, R952	ECO050418-03
134	METAL OXID RESISTOR	10	2W 5% ST BULK			MAN	1	R208	
135	RESISTOR SEMI	100KB	PH0630NOX0X-HA1-B104	SOUNDWELL		MAN	2	RV571, RV572	
136	RESISTOR SEMI	2.2KB	PH0630NOX0X-HA1-B222	SOUNDWELL		MAN	1	RV231	
137	RESISTOR SEMI	22KB	PH0630NOX0X-HA1-B223	SOUNDWELL		MAN	1	RV401	
138	RESISTOR SEMI	4.7KB	PH0630NOX0X-HA1-B472	SOUNDWELL		MAN	2	RV131, RV911	
139	ELECT CAPACITOR	1UF	KR1-16V010MA 5X11	DONGXIANG		MAN	3	C167, C407, C408	
140	ELECT CAPACITOR	10UF	KR1-16V100MA 5X11	DONGXIANG		MAN	7	C232, C331, C332, C333, C913, C921, C931	
141	ELECT CAPACITOR	100UF	KR3-16V101MA 5X11	DONGXIANG		MAN	2	C821, C923	
142	ELECT CAPACITOR	1000UF	KR1-16V102MF 10X20	DONGXIANG		MAN	4	C207, C213, C217, C912	
143	ELECT CAPACITOR	0.22UF	KR1-16VR22MA 5X11	DONGXIANG		MAN	1	C124	
144	ELECT CAPACITOR	2.2UF	KR1-16V2R2MA 5X11	DONGXIANG		MAN	1	C741	
145	ELECT CAPACITOR	22UF	KR1-16V220MA 5X11	DONGXIANG		MAN	1	C221	ECO050412-01
146	ELECT CAPACITOR	220UF	KR1-16V221MB 6.3X20	DONGXIANG		MAN	3	C210, C953, C954	
147	ELECT CAPACITOR	33UF	KR1-16V330MA 5X11	DONGXIANG		MAN	1	C209	
148	ELECT CAPACITOR	4.7UF	KR1-16V4R7MA 5X11	DONGXIANG		MAN	3	C208, C411, C960	
149	ELECT CAPACITOR	47UF	KR1-16V470MA 5X11	DONGXIANG		MAN	5	C143, C185, C206, C233, C933	
150	DISK	100P				MAN	1	C735	
151	DIP TRIMMER	20PF	KCVN620	FOKITS		MAN	1	CT301	ECO050325-04
152	TRANSFORMER	CHOKE	T1	FINE	FINE	MAN	1	T901	ECO050509-02
153	TRANSFORMER	EI-24(MOD.)	T2	FINE	FINE	MAN	1	T201	ECO050509-02
154	IFT COIL RF FREE TX	27MHz	TX1	FINE	FINE	MAN	2	L701, L702	
155	IFT COIL	37MHz TX	TX2	FINE	FINE	MAN	1	L703	
156	IFT COIL	27MHz-B	RX1	FINE	FINE	MAN	1	L101	
157	IFT COIL	27MHz RX (7RC)	RX2	FINE	FINE	MAN	1	L107	
158	IFT COIL	10.7MHz RX-B	RX3	FINE	FINE	MAN	1	L108	
159	IFT COIL DETECTOR	455KHz	RX4	FINE	FINE	MAN	1	L109	
160	IFT COIL	455KHz-B	RX5	FINE	FINE	MAN	1	L112	
161	IFT COIL	16.5MHz VCO	VCO1	FINE	FINE	MAN	1	L301	
162	IFT COIL	15MHz TX/RX	VCO2	FINE	FINE	MAN	1	L302	
163	COIL AXIAL	1.5UH	LAL03TR1R5K			MAN	2	L102, L106	
164	COIL AXIAL	6.8UH	LAL03TB6R8K			MAN	1	L706	
165	COIL SPRING	2.5X0.8X7T:R	0.11UH	FINE	FINE	MAN	1	L704	
166	COIL SPRING	4X0.5X6T:R		FINE	FINE	MAN	1	L712	
167	COIL SPRING	4X0.5X7T:R		FINE	FINE	MAN	1	L713	
168	COIL SPRING	4X0.5X10T:R		FINE	FINE	MAN	1	L714	
169	COIL SPRING	6X0.8X5T:R	0.225UH	FINE	FINE	MAN	1	L711	
170	COIL SPRING	5X0.6X13.5T:R	0.5UH	FINE	FINE	MAN	1	L707	
171	CORE SPRING	5.5TX0.45XM5E@X2.3 0.17UH		FINE	FINE	MAN	3	L103, L104, L105	
172	COIL CHOKE	6UH	BOBBIN CORE	FINE	FINE	MAN	1	L201	
173	TRANSISTOR	KTC1006		KEC		MAN	1	Q703	
174	TRANSISTOR	KTA1241		KEC		MAN	3	Q221, Q951, Q952	ECO050412-01
175	DIODE	1N4004T/R 400V 1A				MAN	2	D232, D901	
176	DIODE	1N60				MAN	1	D161	
177	CRYSTAL	10.240MHz	32PF:30PPM HC-49/U	D.TRON	D.TRON	MAN	1	X101	

ITEM	NAME & DESCRIPTION	VENDOE CODE		MAKER	VENDOR	ASS'Y	Q'ty	Location NO	REMARKS
178	CRYSTAL	4.50MHZ	12PF:30PPM HC-49U	D.TRON	D.TRON	MAN	1	X301	
179	FILTER CRYSTAL	10L08A	10.695MHZ 49/T-3L	D.TRON	D.TRON	MAN	1	XF101	
180	FILTER CERAMIC	LT455HTW		D.TRON	D.TRON	MAN	1	CF101	ECO050525-02
181	JACK MINI	SKJS-206		SEKWANG	SEKWANG	MAN	1	J203	
182	JACK DC	SKJS-3518		SEKWANG	SEKWANG	MAN	1	J202	
183	WAFER	2PIN	PH-2A 180C	FINE	FINE	MAN	1	J205	
184	WAFER	6PIN	PH-6A 180C	FINE	FINE	MAN	1	J503	
185	WAFER	16PIN	PH-16A 180C	FINE	FINE	MAN	1	J102	

Mechanical PARTS

* COVER ASS'Y

1	BOTTOM COVER SPC+BLACK, COATING, 0.75t			Hankyung			1	-	
2	UPPER COVER SPC+BLACK COATING, 0.75t			Hankyung			1	-	
3	BRACKET SPK EGI, 1.0t, COATING	same as TCB880		Hankyung			3	For speaker	
4	FELT(SPK) φ65*0.3t, BLACK			Hankyung			1	For speaker	
5	FELT(COVER) 8*133*0.3t, BLACK			Hankyung			4	For bottom cover & upper cover adhesive(location: side)	
6	SCREW Mach(+) M3*8L-TAPTITE, BLACK, BH	same as TCB880	ECO 050319-02	Bonso			3	Bracket(spk) mounting	
7	SCREW Mach(+) M3*8L-TAPTITE, BLACK, BH	same as TCB880 / same as item 6.		Bonso			6	Main body + Bottom & upper cover mounting	
8	CUSHION 20*24*4.0t, EVA SPONGE, BLACK	same as TCB880	ECO 050328-07	Hankyung			1	For upper cover adhesive(location: center)	

* ESCUTCHEON ASS'Y

9	ESCUTCHEON ABS. BLACK			Bonso			1	Front case plastic	
10	BEZEL ACRYL, CLEAR			Bonso			1	-	
11	DOUBLE TAPE(BEZEL) 3M, 0.16t	ECO 050402-01		Hankyung			1	For Bezel	
12	VOL KNOB ABS, BLACK			Bonso			2	For Rf & On/off	
13	DUAL KNOB ABS, BLACK			Bonso			2	For Rf & On/off	
14	CH KNOB ABS, BLACK	same as TCB880 Vol knob		Bonso			1	For Channel	
15	D-SPRING SK5, 0.2t, Zn plating			TTI			2	For Rf & On/off	
16	KEY PAD SILICONE, CLEAR, PU coating	ECO 050328-07		Hankyung			1	-	
17	LCD BRACKET SPTE, 0.3t			Hankyung			1	For diffuser	
18	DIFFUSER LENS ACRYL, CLEAR			Bonso			1	Diffuser	
19	REFLECTOR ACRYL, CLEAR			Bonso			1	For diffuser	
20	ZEBRA SILICONE			Hankyung			1	For diffuser	
21	WINDOW ACRYL, CLEAR			Bonso			1	-	
22	ILLUMINATION ACRYL, CLEAR			Bonso			1	For Rf & On/off & Channel	
23	SCREW MACH(+) M2.6*5L, Black, FH	same as TCB880		Bonso			4	Esc+Main body mounting	
24	SCREW TAP(+) 2.6*8L-2S, Ni plating, BH			Bonso			5	zpc. illuminator mounting, zpc. key PCB mounting	
25	DOUBLE TAPE(WINDOW) 3M, 0.16t			Hankyung			1	For window	
26	WHITE SHEET POLYESTER, 0.1t			Hankyung			1	For diffuser	
27	RING NUT BsBm	same as TCB880		Hankyung			3	For channel, Rf, on/off	
28	DOUBLE TAPE(BEZEL)-1 23*1.1*0.16T	NEW	ECO 050731-03	Bonso			1	For Bezel	
29	DOUBLE SIDE TAPE FOR DIFFUSER 29*3.5*0.2T	NEW	ECO 050731-03	Bonso			2	For diffuser	

* MAIN BODY ASS'Y

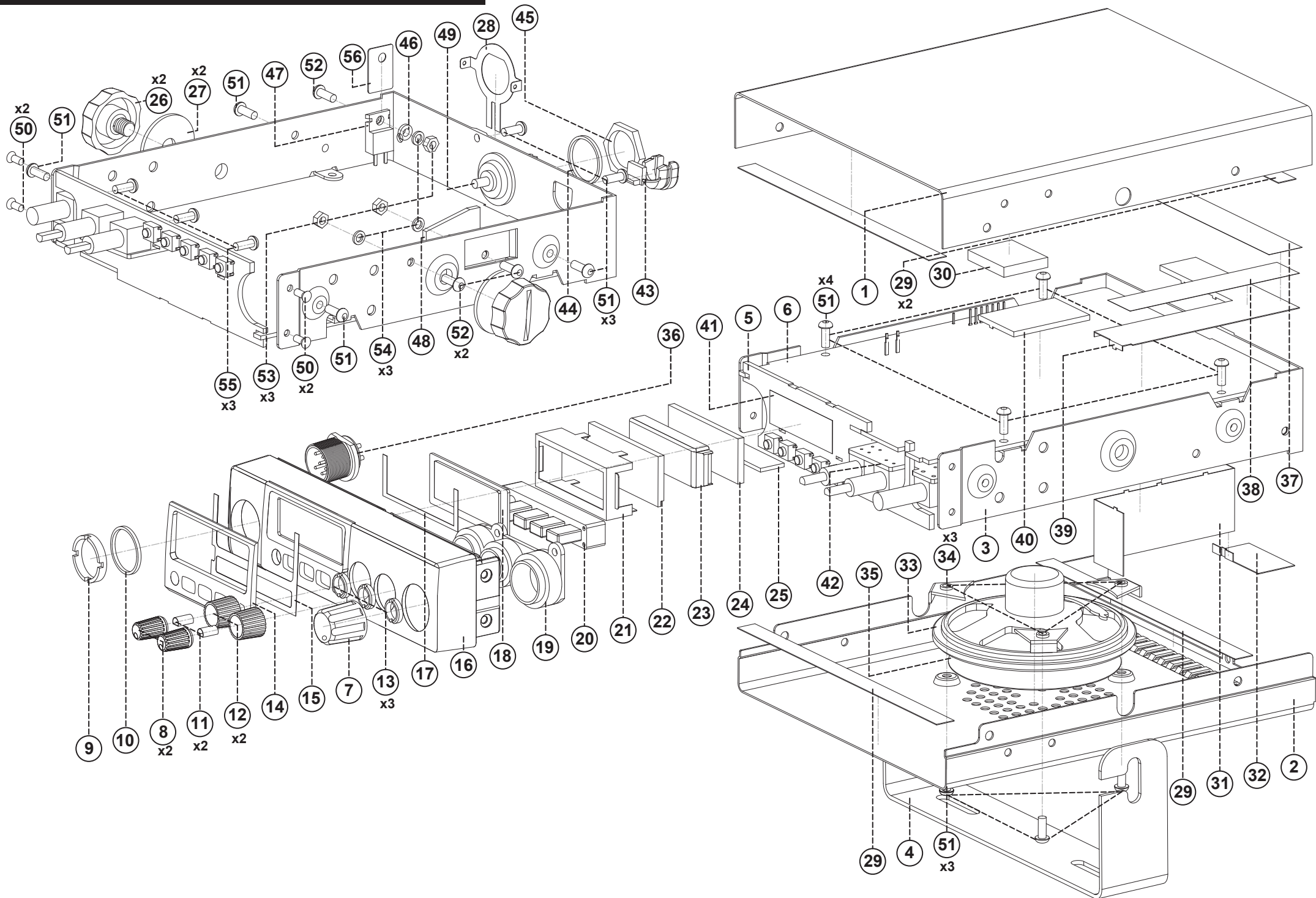
30	MAIN BODY EGI, 1.0t			Hankyung			1	-	
31	CORD STOPPER PP, BLACK	same as TCB880		Hankyung			1	Wire stopper	
32	SCREW MACH(+) M3*10L, Black, BH	same as TCB880	ECO 050412-02	Bonso			3	TR mounting	

ITEM	NAME & DESCRIPTION	VENDOE CODE	MAKER	VENDOR	ASS'Y	Q'ty	Location NO	REMARKS
33	SCREW Mach(+) 3*8L-TAPTITE, Black, BH	same as TCB880 / same as item 6.	Bonso			4	Main PCB + Main body mounting	
34	NUT M3, SS41	same as TCB880	Bonso			3	TR mounting	
35	SPRING WASHER M3	same as TCB880	Bonso			3	TR mounting	
36	HEAT SINK AL, 2.0t	same as TCB880	Hankyung			1	-	
37	BUSH NLLON66+GRASS 15%	same as TCB880	Hankyung			1	For TR	
38	ANT. TERMINAL SPTE, 0.2t	same as TCB880	Hankyung			1	For Antenna	
39	MICA MICA 0.6t	same as TCB880	Hankyung			1	For TR	
40	TX SHIELD CAN(1) SPTE, 0.3t		Hankyung			1	For Main PCB(location: lower)	
41	TX SHIELD CAN(2) SPTE, 0.3t		Hankyung			1	For Main PCB(location: upper)	
42	VCO SHIELD CAN SPTE, 0.3t	same as TCB880	Hankyung			1	For Vco	
43	SHIELD COVER SPTE, 0.3t	ECO 050302-01	Hankyung			1	For TX shield can(2)	
44	Cu Tape-1 Cu 49*9*0.1t	ECO 050328-05	Bonso			1	For TX shield can(1)	
45	Cu Tape-2 Cu 77*15*0.1t	ECO 050328-05 / Same as TCB880	Bonso			1	For TX shield can(1)	

* ACCESSORY								
1	MIC ASS'Y	same as TCB880	Hankyung			1		
2	BRACKET SET		Hankyung			1		
3	STUD BOLT ABS, M6*8L, BLACK	same as TCB880	Hankyung			2		
4	BRACKET(MIC) SUS304, 1.0t	same as TCB880	Hankyung			1		
5	PACKING RUBBER PACKING RUBBER	same as TCB880	Hankyung			2		
6	SCREW Mach(+) M3*8L, Taptite, Black, BH	same as TCB880	Bonso			2		
7	SCREW TAP(+) 5*12L-1S, ZN PLATING, TH	same as TCB880	Bonso			3		
8	STAR WASHER-S M3, NI PLATING, B type	same as TCB880	Bonso			2		
9	STAR WASHER-L M5, NI PLATING, B type	same as TCB880	Bonso			3		
10	POWER CODE	2PIN RECEPRACLE ASS'Y	FINE			1		

* PACKING ASS'Y								
1	OWNER'S MANUAL		Bonso			1		
2	GIFTBOX SW1E 2.0T		Bonso			1		
3	INNER BOX A SW1E 2.0T, BROWN		Bonso			1		
4	INNER BOX B SW1E 2.0T, BROWN		Bonso			1		
5	WARNING LABEL		Bonso			1		
6	CODE 39 SERIAL NO. STICKER		Bonso			2		
7	OUT BOX DW1E 7.0T		Bonso			1		
8	SILICA-GEL 1g		Bonso			1		
9	OPP TAPE 50MM		Bonso			-		

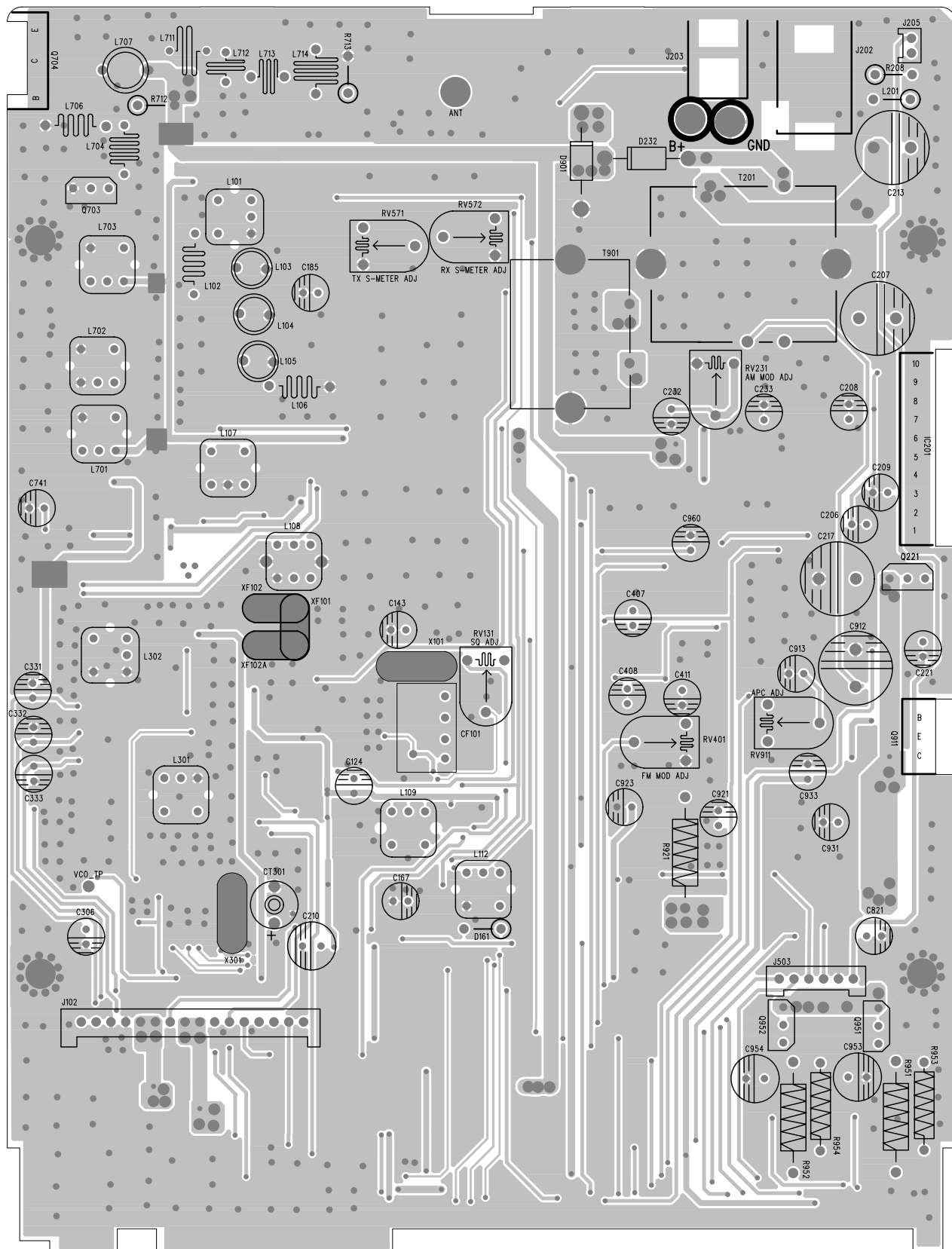
SECTION 4 MECHANICAL DISASSEMBLY



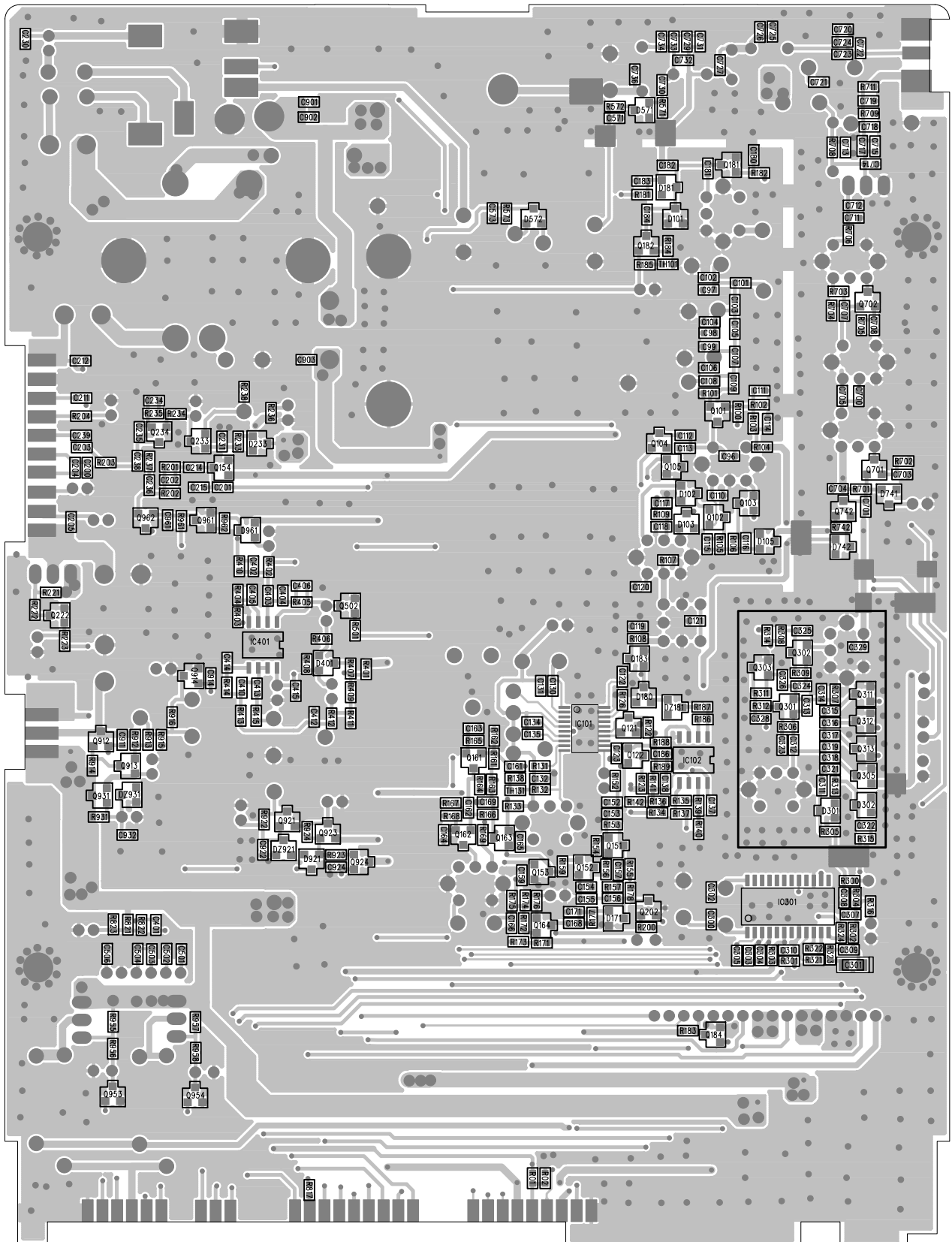
NO	PART NAME	DESCRIPTION	QTY	REMARK
1	Upper cover	SPC 0.75t, Black	1	
2	Bottom cover	SPC 0.75t, Black	1	
3	Main body	EGI, 1.0t	1	
4	Bracket(Set)	SPC 1.6t, Blacke	1	
5	Key PCB	1.6t	1	
6	Main PCB	1.6t	1	
7	Ch knob	ABS, Black□	1	
8	Vol knob	ABS, Black	2	
9	Nut for Mic connector		1	
10	Washer for Mic connector		1	
11	D-Spring	SK5, 0.2t, Zn plating	2	
12	Dual knob	ABS, Black	2	
13	Ring nut	BsBm, Ni plating	3	
14	Bezzel	Acryl	1	
15	Double tape(Bezzel)	0.16t	1	
16	Escutcheon	ABS, Black	1	
17	Double tape(Window)	0.16t	1	
18	Window	Acryl	1	
19	illuminator	Acryl	1	
20	Key pad	Silicone, PU coating	1	
21	LCD Bracket	SPTE, 0.3t	1	
22	LCD		1	
23	Reflector	Acryl	1	
24	Diffuser Lens	Acryl	1	
25	Zebra	Silicone	1	
26	Stud bolt	ABS, BsBm	2	
27	Packing rubber	NBR	2	
28	Ant terminal	SPTC, 0.2t	1	
29	Felt(Cover)	0.3t	4	
30	Cushion	EVA sponge	1	
31	TX shield can(2)	SPTE, 0.3t	1	
32	Shield cover	SPTE, 0.3t	1	
33	Speaker		1	
34	Bracket(Spk)	EGI, 1.0t	3	
35	Felt(Spk)	Dia 65 x 0.3t	1	
36	Mic connector		1	
37	Cu tape-1	Cu, 49x9x0.1t	1	
38	Cu tape-2	Cu, 77x15x0.1t	1	
39	TX shield can(1)	SPTE, 0.3t	1	
40	VCO shield can	SPTE, 0.3t	1	
41	White sheet	PE, 0.1t	1	
42	S/W PCB	PVC	1	
43	Cord stopper		1	
44	Washer for Ant connector		1	
45	Nut for Ant connector		1	
46	Bush	Nylon66xGrass15%	1	For TR
47	TR		2	
48	Heat sink	AL	1	
49	Mic connector		1	
50	Screw Mach(+)	M2.6x5L, FH	4	For Escutcheon
51	Screw Mach(+)	M3x8L_Taptite, BH	13	For Speaker bracket,Main PCB,Cover
52	Screw Mach(+)	M3x10, BH	3	For TR
53	Nut	M3	3	For TR
54	Spring washer	M3	3	For TR
55	Screw Tap(+)	2.6x8L, BH	3	For Key PCB
56	Mica	0.4t	1	

SECTION5 BOARD LAYOUT

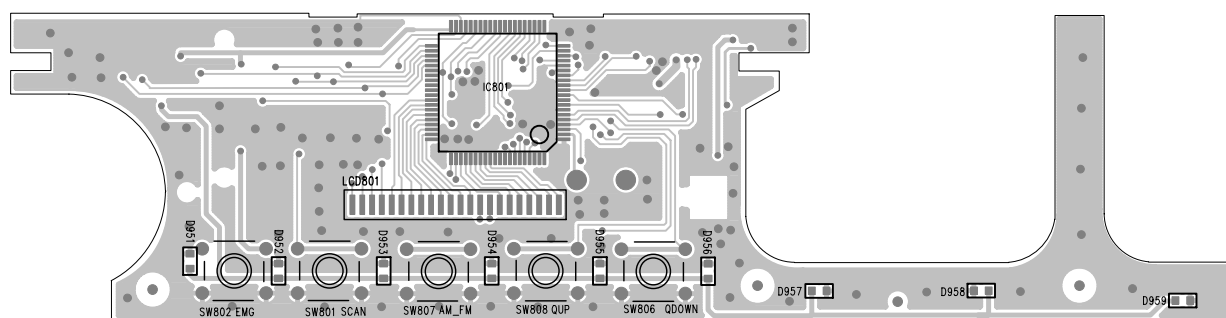
5-1 Main PCB (Top side)



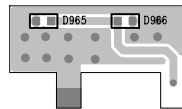
Main PCB (Bottom side)



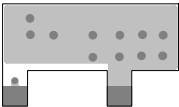
5-2 Front PCB (Top side)



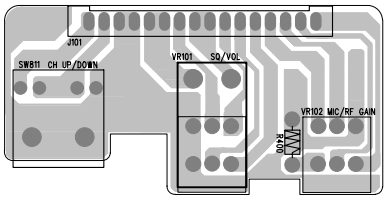
5-3 LED PCB (Top side)



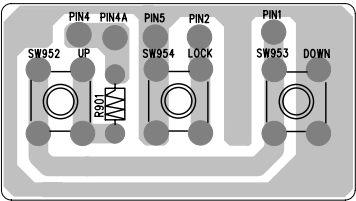
LED PCB (Bottom side)



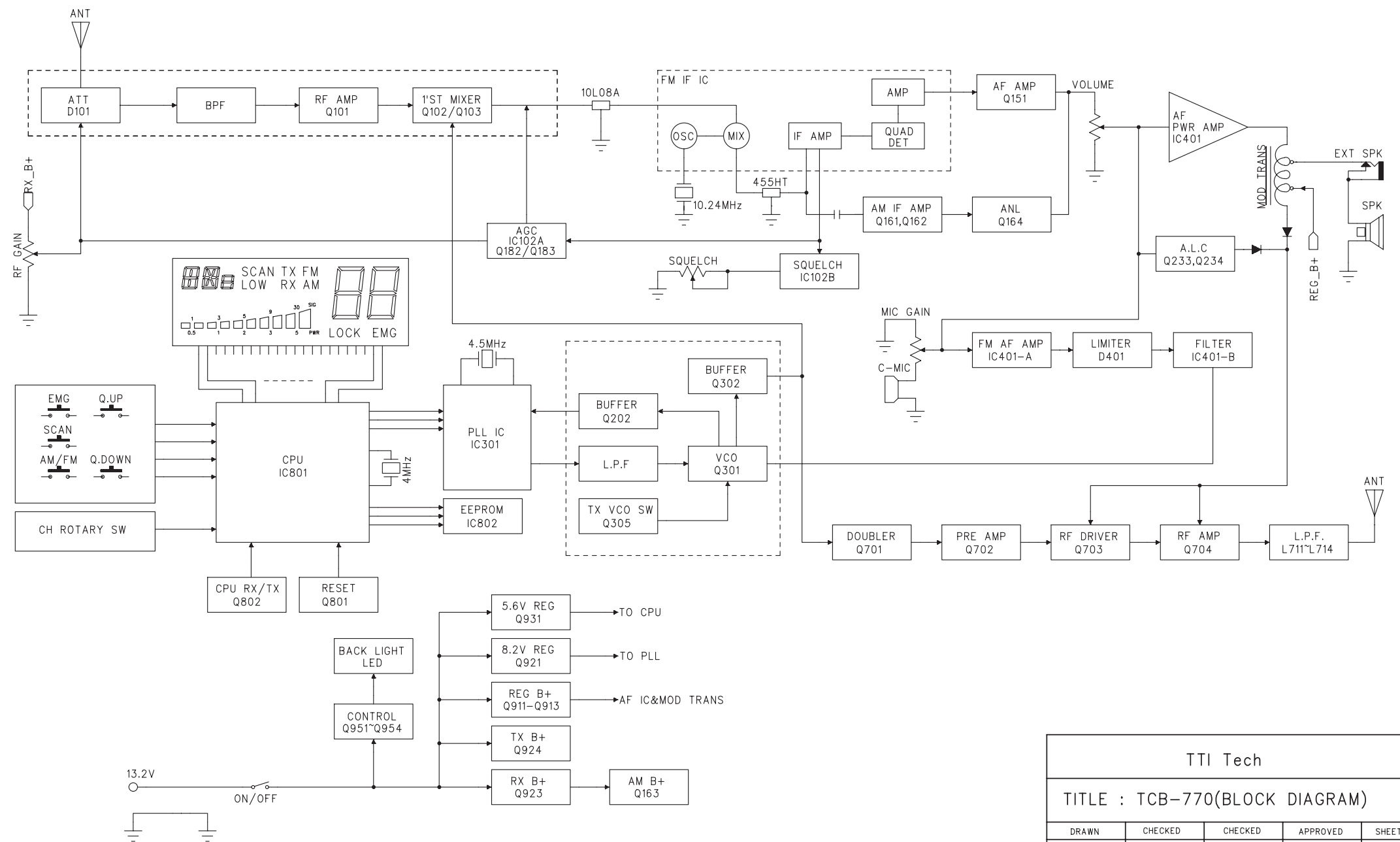
5-4 Volume PCB



5-5 MIC PCB



SECTION6 BLOCK DIAGRAM



TTI Tech				
TITLE : TCB-770(BLOCK DIAGRAM)				
DRAWN	CHECKED	CHECKED	APPROVED	SHEET
S.H.KWON				1 OF 1
				REV
2005.01.31.				0.1

SECTION 7 SCHEMATIC

